

ELAVARASAN DHAYALAN

Senior .NET Developer | 16+ Years of Experience

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PROFESSIONAL SUMMARY

Senior .NET Developer with 16+ years designing and delivering C# applications, enterprise ETL pipelines, and Azure data architectures for Fortune 500 clients including Microsoft and Boeing. Deep expertise in Azure ADF, Cosmos DB, SQL Synapse DW, and modern test automation (Playwright, Selenium), paired with hands-on Python for pipeline monitoring, failure analysis, and predictive analytics and proven expertise in C# ASP.NET, ASP.NET MVC, Web API, and SQL Server. Proven track record leading Agile teams across the US, UAE, UK, and India, mentoring engineers, and shipping production systems on schedule. Actively applies AI-assisted development (GitHub Copilot, Anthropic API) to accelerate delivery and improve code quality.

TECHNICAL SKILLS

Cloud & Data Engineering: Azure ADF, Azure SQL DB, Azure SQL Synapse DW, Cosmos DB, Microsoft Fabric, Kusto (KQL), SSIS, ETL Architecture

Languages & Frameworks: C#, Python, TypeScript, JavaScript, SQL/T-SQL, ASP.NET MVC/Core, Web API, Entity Framework, LINQ, React 15, Angular 8

Test Automation & DevOps: Playwright (TypeScript), Selenium (C#), Azure DevOps Pipelines, CI/CD, Git, PyTest

AI & Analytics: GitHub Copilot, Anthropic API, Scikit-Learn, Pandas, Power BI, SSRS

Other / Legacy Technologies: VB.NET, PowerShell, WCF, MS SQL Server (2000–2022), Git, TFS, ClearCase, Visual SourceSafe

PERSONAL INITIATIVE — FEATURED PROJECT

Personal Finance AI — End-to-End ML Pipeline | Apr 2026

Self-Initiated | Python, Pandas, Scikit-Learn (Random Forest, Linear Regression), joblib, Anthropic API, argparse, python-dotenv, pytest

- Independently designed and built a production-quality personal finance analytics and ML budget prediction engine, extending prior Python scripting experience into full machine learning and data science.
- Built a multi-stage ETL pipeline ingesting 18 months of real transaction data (712 rows) across 3 bank accounts with custom CSV parsers per source format and Pandas-based cleaning (null coercion, deduplication, dtype enforcement, partial-month detection).
- Engineered monthly ML features (lag variables, 3-month rolling averages, z-score anomaly detection) and trained Random Forest and Linear Regression regressors with chronological 80/20 split.
- Refactored into a production layered architecture with dual logging, argparse CLI, joblib model persistence, Anthropic API integration, and 18 pytest unit tests (100% passing) — published at github.com/whoamielavarasan/personal-finance-ai

PROFESSIONAL EXPERIENCE

Adroit Technologies & Cloud Solutions Inc. — Bellevue, WA | Jul 2022 – Present

Computer Systems Analyst

Client: Microsoft | Project: Azure Portal — SQL Synapse DW

Environment: Azure ADF, Cosmos DB, SQL Synapse DW, Scope Script, Kusto, C#, .NET, TypeScript, Playwright, Selenium, Python, Azure DevOps, DacFx/SSDT, SQL Server

- Migrated the full regression test suite from Selenium (C#) to Playwright (TypeScript), significantly reducing transient/flaky test failures and modernizing the automation stack for improved reliability.
- Built Python scripts that proactively analyze test-case failure patterns and generate early-warning alerts, cutting root-cause analysis time and enabling faster bug fixes before deployment.
- Leveraged Azure SRE Agent to streamline failure detection and accelerate root-cause analysis.

- Performed root-cause analysis of production portal failures using Azure DevOps pipeline logs, resolving an average of 5 incidents every two weeks with targeted TypeScript fixes within SLA.
- Designed, coded, and tested Multi-Column Distribution feature in DacFx/SSDT for SQL DW end-to-end using C#.
- Maintained a 110-test-case suite for the SQL Dedicated Pool with full requirement-to-test traceability across local, end-to-end pipeline, and pre-production environments, executed repeatedly across multiple release cycles.
- Mentored 2 junior engineers on TypeScript and Playwright automation, reviewing their work to maintain quality and consistency across the suite.
- Leveraged GitHub Copilot and AI-assisted agents to plan user stories and accelerate bug fixing, noticeably speeding up sprint delivery and improving code consistency.
- Partnered directly with Microsoft client engineers to translate feature requirements into development tasks and acceptance criteria.
- Authored troubleshooting guides and onboarding documentation that streamlined environment setup and ramp-up for new team members.

Quadrant Resources Pvt Ltd — Hyderabad, India | Jul 2021 – Jun 2022

Sr. Technical Analyst

Client: Microsoft | Project: Azure SQL Telemetry — ETL Pipeline Development & Monitoring

Environment: C#, Python, Web API, Azure ADF, Cosmos DB, Azure SQL DB, Scope Script, Kusto, JSON, Power BI, PowerShell

- Developed and maintained ETL workflows using Azure ADF, Cosmos DB, C#, and PowerShell, processing an estimated 1,440+ metric records per database daily (telemetry rows averaging 2–10 KB each) to ensure reliable pipeline operations for Azure SQL Telemetry.
- Wrote Python scripts to proactively analyze Azure ADF pipeline failure patterns, identify root causes, and implement preventive fixes ahead of SLA breaches.
- Owned production support for automated incident management bots that detect, classify, and mitigate pipeline failures, reducing Mean Time to Resolution (MTTR).
- Built Scope scripts to transform raw Cosmos DB data streams (hourly, daily, curated) into structured, curated outputs for downstream reporting and analysis.
- Developed a health-runner monitoring system providing continuous visibility into Azure ADF and Cosmos DB pipeline health.
- Built a Kusto-based logging system and SQL Server metadata store enabling structured telemetry querying and failure trend analysis.
- Delivered an automated C#/stored procedure/Power BI reporting system surfacing ETL lag and gap metrics for real-time pipeline quality dashboards.
- Authored troubleshooting guides and runbooks that standardized incident resolution procedures for new team members.

TechnoCIT Software Solutions — Dubai, UAE | Sep 2016 – Jun 2021

Senior System Analyst · National Archives — Abu Dhabi | Sep 2016 – Aug 2017

Environment: C#, ASP.NET Web API, SQL Server, Tridion 2013, React 15, Selenium

- Designed and built the na.ae public complaint and suggestion portal (React frontend, C# ASP.NET Web API backend), processing 10–100 citizen complaints per month and enabling internal teams to track and resolve them end-to-end.
- Built end-to-end Selenium (C#) automation covering complaint submission, case tracking, and feedback delivery workflows.
- Led full SDLC — requirements, design, development, testing, and production support.
- Cleaned and validated complaints/suggestions data from multiple CSV sources against business rules, then loaded it into structured relational tables.

Senior System Analyst · Mashreq Bank — UAE | Sep 2017 – Jun 2021

Projects: Digital Onboarding · MashreqBank.com · Salaam Rewards · Mashreq Neo

Environment: C#, ASP.NET MVC, WCF, SQL Server, React, Tridion 2013

- Architected and led end-to-end development of Mashreq Neo, a fully branchless digital banking platform for UAE's mobile-first generation.

- Built the Digital Onboarding platform enabling instant account opening, credit card applications, and personal loan requests across mobile, tablet, and web.
- Developed and maintained MashreqBank.com with bilingual (English/Arabic) content management across 15 countries spanning MENA, APAC, Europe, and North America.
- Built Mashreq Salaam Rewards, a banking loyalty portal, using ASP.NET, React, and SQL Server — full feature development and production support.
- Led Agile Scrum ceremonies across 4 concurrent projects, delivering releases on a biweekly cadence.

Senior System Analyst · ADCB Bank — UAE (concurrent) | Jan 2020 – Jun 2021

Environment: C#, ASP.NET, WCF, SQL Server, Tridion 2013, Angular 8

- Developed and enhanced ADCB Bank's regional web portal on Tridion CMS with bilingual (English/Arabic) content management across all Middle East regions.
- Delivered feature enhancements and production support in a high-availability banking environment.

Blackstone Group Technologies — Chennai, India | Jul 2015 – Sep 2016

Consultant

Project: Tekla 3D Modeling Automation

Environment: C#.NET, SQL Server 2008, Visual Studio 2012, Windows Application

- Designed and built a C# Windows application automating 3D modeling workflows in TEKLA, reducing manual modeling effort for engineering teams.
- Architected the application, developed stored procedures, and mentored junior team members.

Impact Standard Business Reporting Pvt Ltd — Coimbatore, India | Dec 2013 – Mar 2015

Tech Lead

Client: GovReports (govreports.com.au) | Projects: GovReports Tax Portal · SuperStream

Environment: C#.NET, SQL Server 2012, MVC5, jQuery, XBRL, WCF, WSE3.0, SVN

- Led architecture and development of GovReports, a multi-tenant Australian tax portal (ITR, CTR, PAYG Summary, BAS, OSR, SuperStream XBRL forms), serving businesses and individuals across Australia.
- Researched SAML security, B2G processes, WCF security, and ebMS3.0/AS4 standards for secure government-to-business data exchange.
- Led requirements, migration analysis, and architecture for the 2015 tax season, planning ebMS3.0/AS4 rollout across all tax forms.
- Delivered SuperStream — an SBR portal with XBRL capabilities for Australia's government e-commerce data standard initiative.

Continental Data Graphics – A Boeing Company — Chennai, India | Apr 2010 – Dec 2013

Software Engineer IV

Client: Boeing | Project: IATP — International Airlines Technical Pool

Environment: C#.NET, ASP.NET MVC, SQL Server 2008, Selenium, ClearCase, SSRS

- Led migration of a legacy ASP application to ASP.NET for the IATP membership platform (Line Maintenance, AOG Aircraft Recovery, Ground Support Equipment inventory), supporting 142 full member airlines and 50 sponsored members.
- Architected the ASP.NET solution and executed DB migration plans with full migration/design documentation.
- Aggregated, cleaned, and validated operational data from multiple airline sources into a centralized database against aviation industry standards.
- Built SSRS reports delivering structured operational data to management for decision-making.
- Ran end-to-end Selenium (C#) automation across all 8 application modules, validating migration quality and regression coverage.
- Provided on-site deployment support at Boeing Long Beach, CA, coordinating go-live and post-deployment production support.

Verilog Networks Pvt Ltd — Chennai, India | Mar 2009 – Apr 2010

Senior Software Engineer

Project: EAPPS — Supply Chain Management

Environment: C#.NET, ASP.NET, SQL Server 2005, VSS

- Designed and developed supply chain ERP modules (Sales, Production, Purchase, Stores, Dispatch, Invoicing, Excise, Payroll, MIS/Analysis Reports) using C#.NET and SQL Server.
- Owned database design, supply chain product analysis, and full development lifecycle delivery.

Wipro Technologies — Hyderabad, India / Norwich, UK | Dec 2005 – Feb 2009

Module Leader · Union Bank of Switzerland — London, UK | Dec 2005 – Feb 2007

Environment: VB.NET, MS SQL Server, ClearCase, Mercury Test Director, DTS, VBA, VBScript

- Designed and developed Business Layer components in VB.NET for a Risk Ratings platform (ratings workflow, override management, static data maintenance), delivered defect-free to UBS London.
- Built a DTS-based ETL migration from MS Access to SQL Server, ingesting feeds from 4 source systems (TLM, GCU, HARD, GEARS) with automated error logging.

Module Leader – Onsite · Aviva Insurance — Norwich, UK | Mar 2007 – Jun 2008

Environment: SQL Server 2000, DTS, VBA, VBScript, C#, ASP.NET, QTP, Mercury Test Director, CVS

- Designed a DTS-driven automation system replacing manual claims tracking, automatically routing new business data from Tranzax to the business team.
- Built an ETL feed pipeline standardizing Bodily Injury claims data from multiple systems into a unified MI file structure; wrote QTP (VBScript) regression tests for the pipeline.
- Served as onsite module lead for the BDC module and Build Manager, coordinating an offshore delivery team.

Senior Software Engineer · Harper Collins | Jul 2008 – Feb 2009

Environment: C#.NET, ASP.NET, SQL Server 2005, FAST Search, VSS

- Built eSearch, a dictionary search engine, using FAST Search technology — covering content ingestion, transformation, relevancy analysis, indexing, and query processing.

EARLY CAREER EXPERIENCE (1997 – 2005)

Blackstone Group Technologies — Chennai, India | Sep 2004 – Dec 2005

Senior Software Engineer. Designed and developed a 3-tier ERP system (Order Processing, Design, Project, Workshop, Store, Finance, Payroll) and a manufacturing cost-calculation tool using VB.NET, Web Services, and SQL Server.

Future Focus InfoTech Pvt. Ltd. — Chennai, India | Jul 1999 – Sep 2004

Software Engineer. Delivered payroll, subcontract management, and service management systems for enterprise clients using VB6, ASP, SQL Server, MySQL, and Crystal Reports.

Chennai Interactive Business Services P Ltd — Chennai, India | Aug 1998 – Jul 1999

Faculty cum Software Engineer. Delivered training programs in ASP, VBScript, and HTML for students and staff.

Aptech Computer Education — Chennai, India | Aug 1997 – Jul 1998

Faculty. Provided technical training in MS Office, FoxPro, VBScript, and HTML fundamentals.

EDUCATION

Bachelor of Engineering (B.E.) — Computer Science & Engineering

Bharathidasan University, India · 1998